

Fraud Detection System Diagram Specs

FraudWatch AI Engineering Specifications • Generated Document

FraudWatch AI: Fraud Detection System Diagram Specifications

This document defines the complete technical specifications, component breakdown, levels, features, and notations required to build Data Flow Diagrams (DFD - Levels 0 to 3), Use Case Diagrams, Entity-Relationship (ER) Diagrams, and Activity Diagrams tailored specifically for the FraudWatch AI Financial Fraud Detection & Risk Analyzer system.

1. System Overview & Component Mapping

Component | Functionality
Ingestion Engine | Upload CSV, XLSX, PDF (<10MB); parse, deduplicate, missing cell imputation, Z-Score normalization.
ML Ensemble Engine | Parallel execution of 4 unsupervised models: Isolation Forest (35%), AutoEncoder (30%), LOF (20%), One-Class SVM (15%).
GenAI Threat Analyzer | Integrates Google Gemini 2.5 Flash API for natural language risk explanations and interactive Q&A assistant.
Export & Reporting | PDFKit-based PDF report builder generating downloadable audit packages.
Data Layer | PostgreSQL database with Drizzle ORM managing dataset metadata, flagged transactions, users, and audit logs.

2. Multi-Level Data Flow Diagram (DFD) Specifications

2.1 Level 0: Context Diagram (System Boundary)

The Context Diagram encapsulates the entire FraudWatch AI System into a single central Process Bubble (Process 0) surrounded by External Entities.

```
graph TD
    Analyst["External Entity: Financial Analyst"]
    Admin["External Entity: Platform Administrator"]
    GeminiAPI["External Entity: Google Gemini GenAI API"]

    Analyst -->| "1. Raw Transaction Dataset (CSV/XLSX/PDF), Q&A Prompts" | Process0(("0.0 FraudWatch AI Fraud Detection System"))
    Process0 -->| "2. Risk Dashboard, Flagged Anomaly Explanations, PDF Reports" | Analyst

    Admin -->| "3. System Configuration & User Credentials" | Process0
    Process0 -->| "4. System Audit Logs & Global Fraud Metrics" | Admin

    Process0 -->| "5. Transaction Feature Summary & Threat Prompts" | GeminiAPI
    GeminiAPI -->| "6. Executive Threat Summaries & Explanation Text" | Process0
```

Elements & Features (Level 0):

- Process Bubble: `0.0 FraudWatch AI Fraud Detection System`
- External Entities:
 1. Financial Analyst: Uploads files, configures threshold parameters, views anomaly tables, triggers GenAI reports, downloads PDFs.
 2. Platform Administrator: Monitors system health, manages user permissions, views global system metrics and audit trails.
 3. Google Gemini GenAI API: External REST endpoint receiving aggregated anomaly features and returning explanation text.
 - Level 0 Data Flows: Raw Dataset Input, Flagged Anomaly Output, AI Prompt Payload, AI Explanation Response, Audit Log Data.

2.2 Level 1: System Overview DFD

Decomposes Process 0 into 5 core system sub-processes and introduces 5 primary Data Stores.

```
graph TD
    Analyst["Financial Analyst"]
    Admin["Platform Administrator"]
    Gemini["Gemini GenAI API"]

    D1["D1: Raw Dataset Store"]
    D2["D2: Preprocessed Features Store"]
    D3["D3: Anomaly & Risk Score Store"]
    D4["D4: AI Summaries & Reports Store"]
    D5["D5: Users & System Audit Logs"]

    P1["1.0 File Ingestion & Data Preprocessing"]
    P2["2.0 ML Unsupervised Ensemble Execution"]
    P3["3.0 GenAI Threat & Risk Explainer"]
    P4["4.0 Report Generation & PDF Exporter"]
    P5["5.0 User Access & System Audit Logging"]

    Analyst -->|"Upload CSV/XLSX/PDF"| P1
    P1 -->|"Store Raw Upload"| D1
    P1 -->|"Cleaned & Scaled Data"| D2

    D2 -->|"Feature Matrices"| P2
    P2 -->|"Weighted Anomaly Scores"| D3

    D3 -->|"Flagged High-Risk Transactions"| P3
    P3 <-->|"Prompt & Response Payload"| Gemini
    P3 -->|"Natural Language Explanations"| D4

    D3 & D4 -->|"Fetch Anomaly & Summary Data"| P4
    P4 -->|"Formatted PDF Package"| Analyst

    Admin -->|"Manage System Users"| P5
    P5 <-->|"User Roles & History Logs"| D5
```

Level 1 Sub-Processes:

1. `1.0 File Ingestion & Data Preprocessing`: Validates, cleans, imputes, and normalizes input data.
2. `2.0 ML Unsupervised Ensemble Execution`: Computes anomaly scores using Isolation Forest, LOF, SVM, and AutoEncoder.
3. `3.0 GenAI Threat & Risk Explainer`: Formulates prompts and queries Gemini API for text explanations and chat responses.
4. `4.0 Report Generation & PDF Exporter`: Assembles dataset metadata, graphs, and AI text into PDF files.
5. `5.0 User Access & System Audit Logging`: Validates session tokens, enforces RBAC, and logs system events.

Level 1 Data Stores:

- `D1: Raw Dataset Store` (Uploaded files & raw records)
- `D2: Preprocessed Features Store` (Scaled Z-scores, cleaned features)
- `D3: Anomaly & Risk Score Store` (Model scores, weighted final score, risk levels: High/Medium/Low)
- `D4: AI Summaries & Reports Store` (Gemini executive summaries, transaction explanations, chat history)
- `D5: Users & System Audit Logs` (User credentials, roles, login history, API logs)

2.3 Level 2: Sub-Process Decomposition DFDs

Level 2 DFD — Decomposition of Process 2.0 (ML Ensemble Engine)

```
graph TD
    D2["D2: Preprocessed Features"]
    D3["D3: Anomaly Score Store"]

    P21["2.1 Parallel Model Dispatcher"]
    P22["2.2 Isolation Forest Execution (35%)"]
    P23["2.3 AutoEncoder Execution (30%)"]
    P24["2.4 Local Outlier Factor Execution (20%)"]
    P25["2.5 One-Class SVM Execution (15%)"]
    P26["2.6 Weighted Majority Voting & Risk Categorizer"]

    D2 -->|"Normalized Matrix"| P21
    P21 -->|"Input Vector"| P22 & P23 & P24 & P25

    P22 -->|"IF Score [0..1]"| P26
    P23 -->|"AE Score [0..1]"| P26
    P24 -->|"LOF Score [0..1]"| P26
    P25 -->|"SVM Score [0..1]"| P26

    P26 -->|"Composite Risk Score & Risk Tag (HIGH / MED / LOW)"| D3
```

Features & Level Balancing Rules for Process 2.0:

- Parent Input: Cleaned features from `D2`.
- Parent Output: Anomaly scores & risk tags written to `D3`.
- Sub-processes:
- `2.1 Dispatcher`: Receives matrix, spawns 4 parallel worker threads/promises.
- `2.2 - 2.5 Individual ML Algorithms`: Compute normalized outlier probability `S\_i` " [0, 1]`.
- `2.6 Ensemble Aggregator`: Calculates $Score_{final} = 0.35(IF) + 0.30(AE) + 0.20(LOF) + 0.15(SVM)$. Categorizes score into High (≥ 0.75), Medium ($0.45 - 0.74$), Low (< 0.45).

2.4 Level 3: Primitive DFD & Process Specifications (P-Specs)

Process Specification P-Spec 2.6: Weighted Majority Voting Algorithm

```
ALGORITHM: CalculateWeightedEnsembleScore
INPUT: S_IF (float), S_AE (float), S_LOF (float), S_SVM (float)
OUTPUT: FinalScore (float), RiskCategory (string)

BEGIN
    FinalScore = (0.35 * S_IF) + (0.30 * S_AE) + (0.20 * S_LOF) + (0.15 * S_SVM)

    IF FinalScore >= 0.75 THEN
        RiskCategory = "HIGH_RISK"
    ELSE IF FinalScore >= 0.45 THEN
        RiskCategory = "MEDIUM_RISK"
    ELSE
        RiskCategory = "LOW_RISK"
    ENDIF

    RETURN FinalScore, RiskCategory
END
```

3. Use Case Diagram Specifications for FraudWatch

3.1 Actors & System Boundary

- Primary Actor 1: `Financial Analyst` (Standard User — uploads data, reviews alerts, queries AI).
- Primary Actor 2: `Platform Administrator` (System Owner — views audit logs, manages system users).
- Secondary Actor (External System): `Google Gemini GenAI Engine` (Processes prompts, generates NLP explanations).

3.2 Use Case Diagram Visualization

```
graph LR
  Analyst(("Financial Analyst"))
  Admin(("Platform Admin"))
  Gemini(("Google Gemini API"))

  subgraph FraudWatch System Boundary
    UC1(["UC-1: Authenticate User"])
    UC2(["UC-2: Upload Transaction Dataset"])
    UC3(["UC-3: Preprocess & Normalize Data"])
    UC4(["UC-4: Run ML Ensemble Engine"])
    UC5(["UC-5: View Anomaly Dashboard"])
    UC6(["UC-6: Generate GenAI Explanations"])
    UC7(["UC-7: Interactive AI Fraud Chat"])
    UC8(["UC-8: Export PDF Audit Report"])
    UC9(["UC-9: View System Audit Logs"])
    UC10(["UC-10: Manage User Accounts"])
  end

  Analyst --> UC1
  Analyst --> UC2
  Analyst --> UC5
  Analyst --> UC7
  Analyst --> UC8

  Admin --> UC1
  Admin --> UC9
  Admin --> UC10

  UC2 .->|"«include»"| UC3
  UC3 .->|"«include»"| UC4
  UC5 .->|"«extend»"| UC6

  UC6 --> Gemini
  UC7 --> Gemini
```

3.3 Detailed Use Case Specifications Table

Use Case ID	Name	Primary Actor	Included / Extending Use Cases	Description
UC-1	Authenticate User	Analyst / Admin	—	Logs user in via secure cookie session and validates role (RBAC).
UC-2	Upload Transaction Dataset	Financial Analyst	Includes UC-3	Ingests CSV, XLSX, or PDF dataset (<10MB).
UC-3	Preprocess & Normalize Data	System (Internal)	Includes UC-4	Performs deduplication, null imputation, and Z-score scaling.
UC-4	Run ML Ensemble Engine	System (Internal)	—	Runs IF, AE, LOF, and SVM in parallel; computes weighted anomaly scores.
UC-5	View Anomaly Dashboard	Financial Analyst	Extended by UC-6	Displays interactive table of transactions sorted by risk score.
UC-6	Generate GenAI Explanations	Financial Analyst	Calls Gemini API	Requests natural language explanation of why specific transactions were flagged.
UC-7	Interactive AI Fraud Chat	Financial Analyst	Calls Gemini API	Enables interactive natural language Q&A regarding dataset patterns.
UC-8	Export PDF Audit Report	Financial Analyst	—	Downloads custom PDF report containing summaries, charts, and flagged rows.
UC-9	View System Audit Logs	Platform Admin	—	Displays system operation history, upload logs, and user activity.
UC-10	Manage User Accounts	Platform Admin	—	Admin creates, deactivates, or updates user permissions and roles.

4. Entity-Relationship (ER) Diagram Specifications

4.1 Entities & Relational Schema (Crow's Foot Notation)

```
erDiagram
    USERS ||--o{ DATASETS : uploads
    USERS ||--o{ AUDIT_LOGS : generates
    DATASETS ||--|{ TRANSACTIONS : contains
    DATASETS ||--o{ AI_SUMMARIES : produces
    DATASETS ||--o{ CHAT_MESSAGES : contextualizes
    TRANSACTIONS ||--|{ ANOMALY_RESULTS : evaluates

    USERS {
        uuid id PK
        string email UK
        string password_hash
        string name
        string role "ANALYST | ADMIN"
        timestamp created_at
    }

    DATASETS {
        uuid id PK
        uuid user_id FK
        string file_name
        string file_type "CSV | XLSX | PDF"
        int total_records
        int flagged_anomalies
        timestamp uploaded_at
    }

    TRANSACTIONS {
        uuid id PK
        uuid dataset_id FK
        string transaction_id_ext
        decimal amount
        string timestamp_raw
        string merchant_category
        jsonb custom_features
    }

    ANOMALY_RESULTS {
        uuid id PK
        uuid transaction_id FK
        float score_isolation_forest
        float score_autoencoder
        float score_lof
        float score_ocsvm
        float final_weighted_score
        string risk_level "HIGH | MEDIUM | LOW"
    }
```

```
AI_SUMMARIES {  
  uuid id PK  
  uuid dataset_id FK  
  text executive_summary  
  jsonb anomaly_explanations  
  timestamp generated_at  
}
```

```
CHAT_MESSAGES {  
  uuid id PK  
  uuid dataset_id FK  
  uuid user_id FK  
  string sender "USER | AI"  
  text message_text  
  timestamp timestamp  
}
```

```
AUDIT_LOGS {  
  uuid id PK  
  uuid user_id FK  
  string action_type  
  string ip_address  
  text details  
  timestamp timestamp  
}
```

4.2 Entity Field Specification Table

1. `USERS`: Manages analyst and admin login profiles.
2. `DATASETS`: Metadata for uploaded CSV/XLSX/PDF files (`user\_id` FK to `USERS`).
3. `TRANSACTIONS`: Parsed transaction records belonging to a dataset (`dataset\_id` FK to `DATASETS`).
4. `ANOMALY\_RESULTS`: Individual model breakdown and weighted final risk score for each transaction (`transaction\_id` FK to `TRANSACTIONS`).
5. `AI\_SUMMARIES`: Gemini-generated audit summary text and bullet points for a dataset (`dataset\_id` FK to `DATASETS`).
6. `CHAT\_MESSAGES`: Stores conversational AI Q&A history per user/dataset session.
7. `AUDIT\_LOGS`: Records administrative and security events for compliance monitoring.

5. Activity Diagram Specifications for FraudWatch

5.1 End-to-End Fraud Detection Workflow

This Activity Diagram models the multi-threaded execution from file upload to anomaly detection, GenAI enrichment, and PDF report export, partitioned across Swimlanes.

```
stateDiagram-v2
    state "1. Financial Analyst (UI)" as UI {
        [*] --> UploadFile
        UploadFile --> SelectThreshold
        ViewResults --> ClickExplain
        ClickExplain --> ViewAIExplanation
        ViewAIExplanation --> ClickExportPDF
        ClickExportPDF --> DownloadPDF
        DownloadPDF --> [*]
    }

    state "2. Next.js Server & Preprocessor" as Server {
        UploadFile --> ValidateFormat

        state FormatCheck <<choice>>
        ValidateFormat --> FormatCheck
        FormatCheck --> ParseAndClean: Valid Format (<10MB)
        FormatCheck --> ReturnError: Invalid File / Too Large
        ReturnError --> UploadFile

        ParseAndClean --> ScalingZScore
    }

    state "3. ML Ensemble Worker" as MLEngine {
        state ForkML <<fork>>
        ScalingZScore --> ForkML

        ForkML --> RunIF
        ForkML --> RunAE
        ForkML --> RunLOF
        ForkML --> RunSVM

        state JoinML <<join>>
        RunIF --> JoinML
        RunAE --> JoinML
        RunLOF --> JoinML
        RunSVM --> JoinML

        JoinML --> ComputeWeightedScore
        ComputeWeightedScore --> CategorizeRisk
        CategorizeRisk --> SaveToDB
    }

    state "4. Gemini GenAI Service" as GenAI {
        ClickExplain --> ConstructPrompt
        ConstructPrompt --> CallGeminiAPI
        CallGeminiAPI --> ReturnExplanation
        ReturnExplanation --> ViewAIExplanation
    }

    state "5. PDF Generator Component" as PDF {
        ClickExportPDF --> AssembleData
        SaveToDB --> ViewResults
        AssembleData --> RenderPDFKit
        RenderPDFKit --> DownloadPDF
    }
```

}

5.2 Key Control Flow Highlights:

1. Validation Decision Node: Checks format (CSV/XLSX/PDF) and file size limitation (<10MB).
2. Parallel Processing (Fork Bar): Triggers Isolation Forest, AutoEncoder, LOF, and One-Class SVM simultaneously to ensure sub-5-second execution performance.
3. Synchronization (Join Bar): Waits for all 4 ML algorithms to finish before computing the combined weighted score.

4. Conditional GenAI Route: GenAI explanations are fetched on-demand or automatically for high-risk transactions.

6. Diagram Implementation & Rendering Summary Matrix

Diagram Type | FraudWatch Component Represented | Key Structural Elements | SQL / Code Artifact Generated

Context DFD (Level 0) | Entire System Boundary | Process 0, Analyst, Admin, Gemini API | Architecture overview diagram

System DFD (Level 1) | Core Next.js Pipeline & Stores | 5 Processes (1.0 to 5.0), 5 Data Stores (D1-D5) | High-level data pipeline schema

Detailed DFD (Level 2) | ML Ensemble Engine (Process 2.0) | Dispatcher, IF, AE, LOF, SVM, Weighted Aggregator | Model execution flow & data contracts

Primitive DFD (Level 3) | Weighted Voting Logic | Pseudocode & Process Specifications (P-Specs) | `calculateEnsembleScore()` function

Use Case Diagram | User Roles & System Features | Actors (Analyst, Admin, Gemini), 10 Use Cases | RBAC matrix & API endpoint mapping

ER Diagram | PostgreSQL Database Schema | 7 Entities (`users`, `datasets`, `transactions`, etc.) | Drizzle ORM schema (`schema.ts`) / SQL DDL

Activity Diagram | End-to-End Processing Workflow | Swimlanes, Decision Diamond, Fork/Join Bars | Control flow & state transition logic

